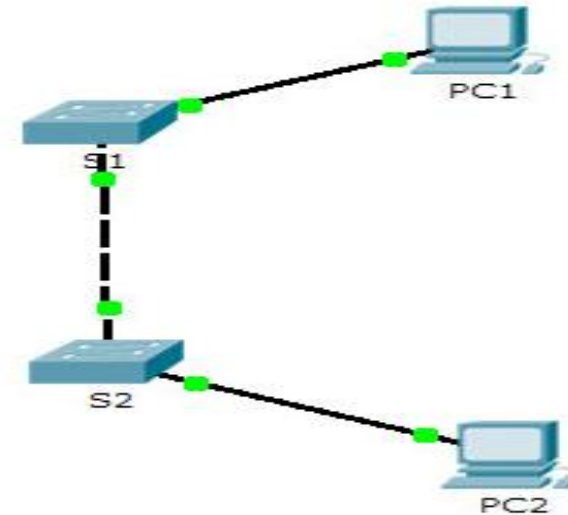
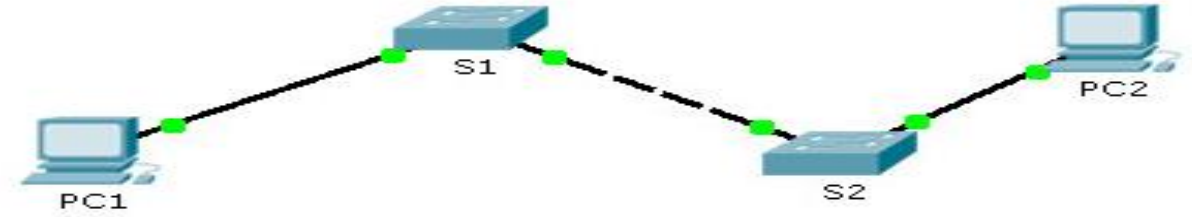


Basic Switch Configuration



Agenda

Cisco IOS

Configure a Cisco Switch

Save a Config

Basic Switch Security

Cisco IOS

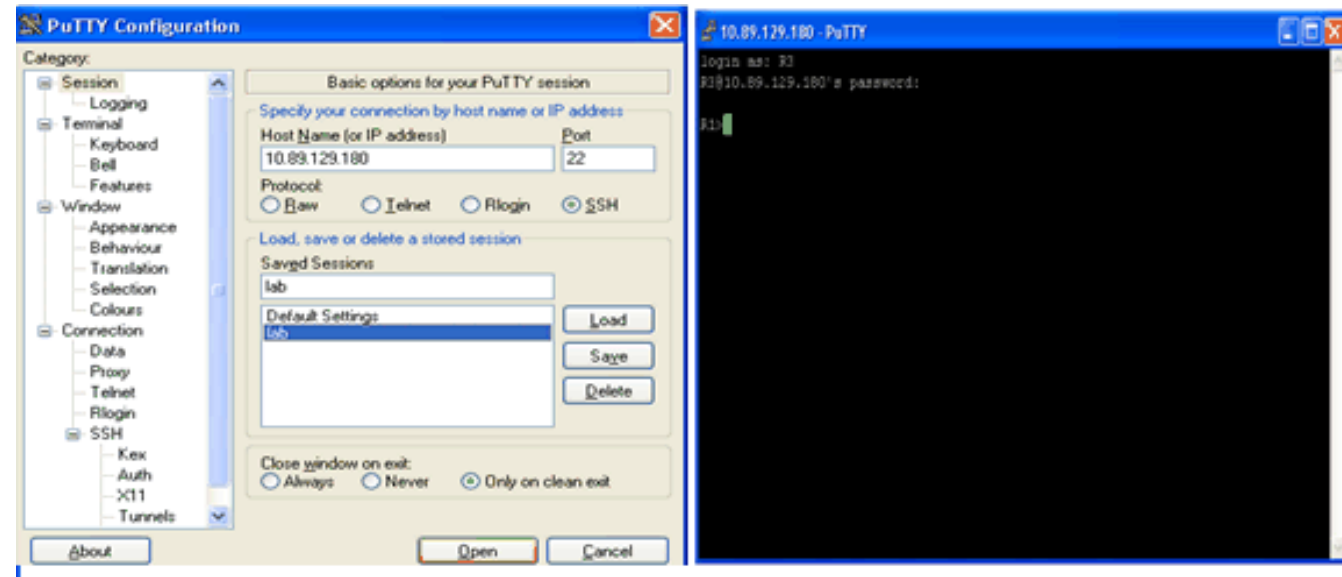
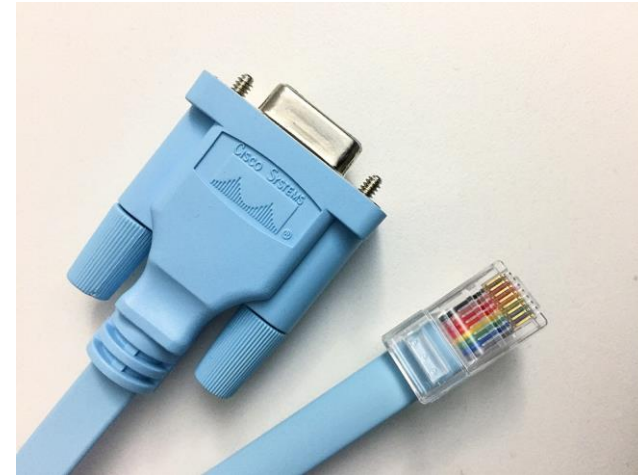
- Cisco Internetwork Operating System (IOS)
- Used on Cisco Systems switches and routers



Networking
Academy

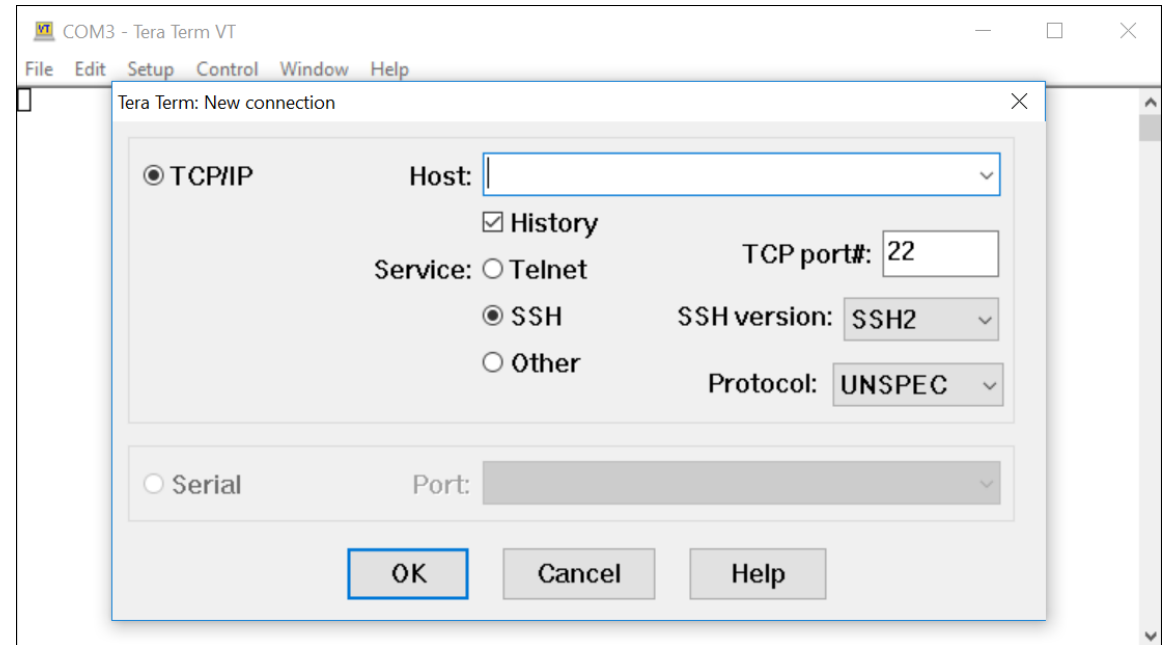
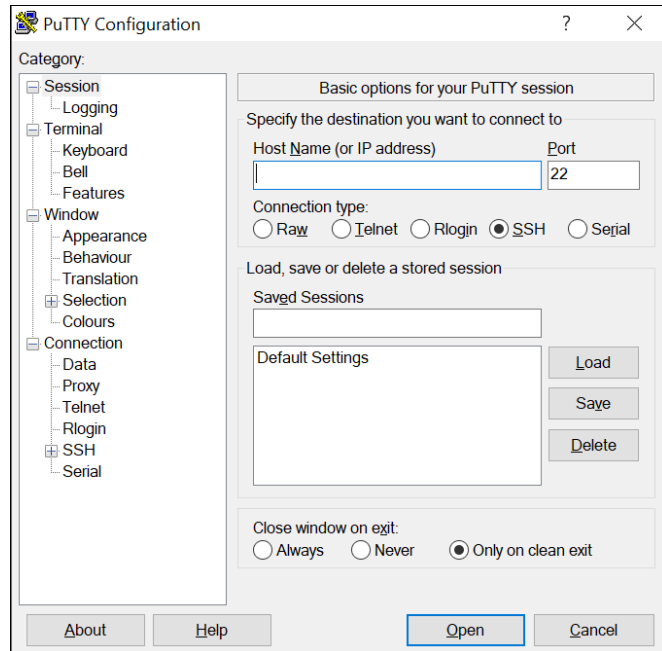
Access Methods

- **Console** – A physical management port used to access a device in order to provide maintenance, such as performing the initial configurations.
- **Secure Shell (SSH)** – Establishes a secure remote CLI connection to a device, through a virtual interface, over a network. (Note: This is the recommended method for remotely connecting to a device.)
- **Telnet** – Establishes an insecure remote CLI connection to a device over the network. (Note: User authentication, passwords and commands are sent over the network in plaintext.)



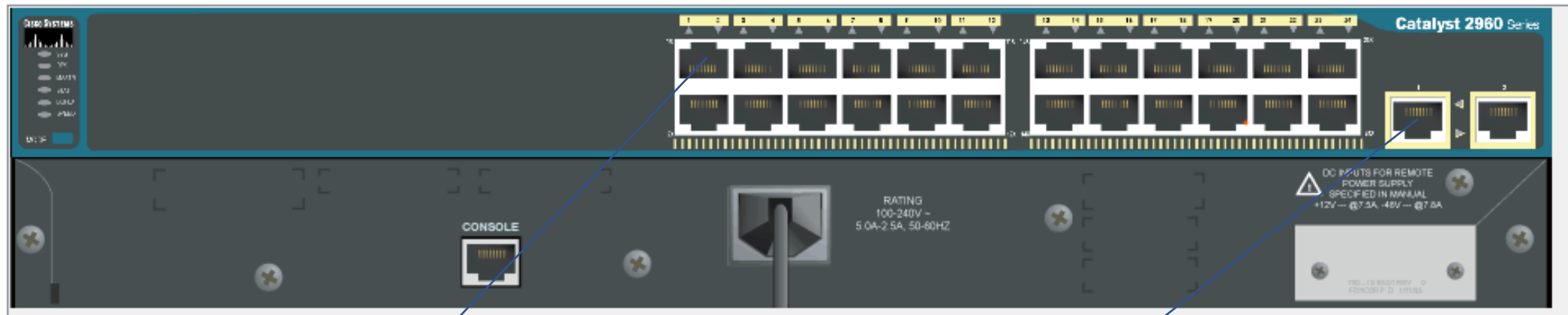
Terminal Emulation Programs

- Terminal emulation programs are used to connect to a network device by either a console port(directly to the device) or by an SSH/Telnet connection(usually over a network).
- There are several terminal emulation programs to chose from such as PuTTY, Tera Term and SecureCRT.



Cisco Device interfaces: Port naming Convention

- Cisco router interfaces are named with the following convention:
 - Media-type slot-number/port-number
OR
 - Media-type slot-number/interface-slot-number/port-number



FastEthernet 0/1
fa0/1

GigabitEthernet 0/1
G0/1

IOS Navigation

Primary Command Modes

User EXEC Mode:

- Allows access to only a limited number of basic monitoring commands
- Identified by the CLI prompt that ends with the > symbol

```
Router>  
  
Switch>
```

Privileged EXEC Mode:

- Allows access to all commands and features
- Identified by the CLI prompt that ends with the # symbol

```
Router#  
  
Switch#
```


Configuration Mode and Subconfiguration Modes

Global Configuration Mode:

- Used to access configuration options on the device

```
Switch(config) #
```

Line Configuration Mode:

- Used to configure console, SSH, Telnet or AUX access

```
Switch(config-line) #
```

Interface Configuration Mode:

- Used to configure a switch port or router interface

```
Switch(config-if) #
```

Navigation Between IOS Modes

- **Privileged EXEC Mode:**

- To move from user EXEC mode to privilege EXEC mode, use the **enable** command.

```
Switch> enable  
Switch#
```

- **Global Configuration Mode:**

- To move in and out of global configuration mode, use the **configure terminal** command. To return to privilege EXEC mode, use the **exit** command.

```
Switch(config)#  
Switch(config)#exit  
Switch#
```

- **Line Configuration Mode:**

- To move in and out of line configuration mode, use the **line** command followed by the management line type. To return to global configuration mode, use the **exit** command.

```
Switch(config)#line console 0  
Switch(config-line)#exit  
Switch(config)#
```

Navigation Between IOS Modes (Cont.)

Subconfiguration Modes:

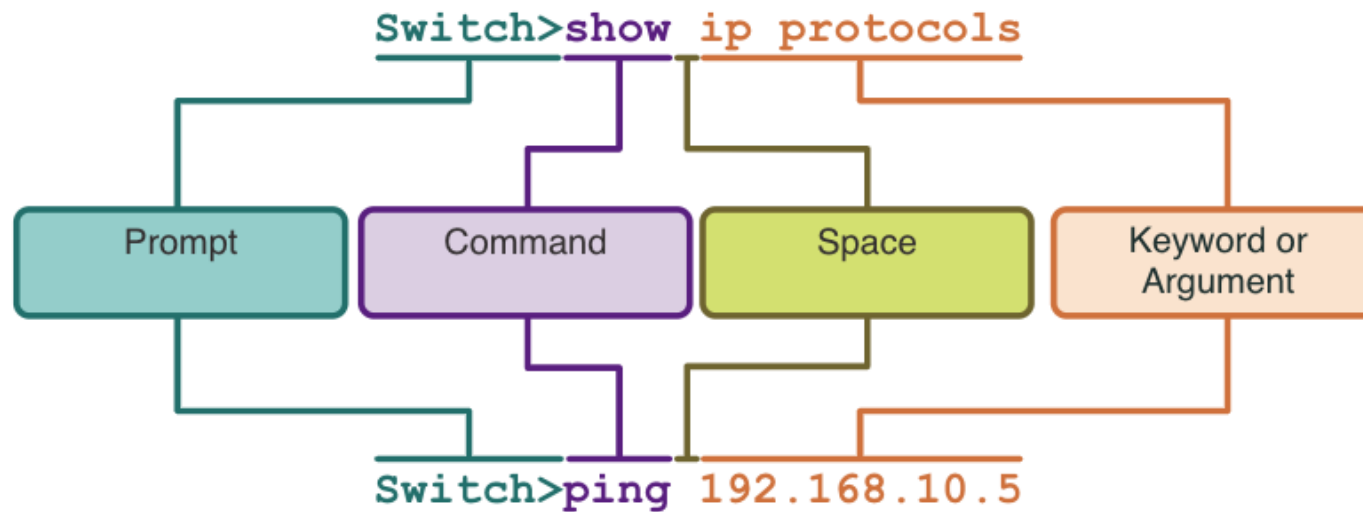
- To move out of any subconfiguration mode to get back to global configuration mode, use the **exit** command. To return to privilege EXEC mode, use the **end** command or key combination **Ctrl +Z**.
- To move directly from one subconfiguration mode to another, type in the desired subconfiguration mode command. In the example, the command prompt changes from **(config-line)#** to **(config-if)#**.

```
Switch(config)#line console 0  
Switch(config-line)#end  
Switch#
```

```
Switch(config-line)#interface FastEthernet 0/1  
Switch(config-if)#
```

IOS Command Structure

Basic IOS Command Structure



- **Keyword** — This is a specific parameter defined in the operating system (in the figure, **ip protocols**).
- **Argument** - This is not predefined; it is a value or variable defined by the user (in the figure, **192.168.10.5**).

IOS Help Features

The IOS has two forms of help available: context-sensitive help and command syntax check.

- Context-sensitive help enables you to quickly find answers to these questions:
 - Which commands are available in each command mode?
 - Which commands start with specific characters or group of characters?
 - Which arguments and keywords are available to particular commands?
- Command syntax check verifies that a valid command was entered by the user.
 - If the interpreter cannot understand the command being entered, it will provide feedback describing what is wrong with the command.

```
Router#ping ?  
WORD  Ping destination address or hostname  
ip     IP echo  
ipv6   IPv6 echo
```

```
Switch#interface fastEthernet 0/1  
                ^  
% Invalid input detected at '^' marker.
```

Hot Keys and Shortcuts

- The IOS CLI provides hot keys and shortcuts that make configuring, monitoring, and troubleshooting easier.
- Commands and keywords can be shortened to the minimum number of characters that identify a unique selection. For example, the **configure** command can be shortened to **conf** because **configure** is the only command that begins with **conf**.

```
Router#con
% Ambiguous command: "con"
Router#con?
configure  connect
```

Keystroke	Description
Tab	Completes a partial command name entry.
Backspace	Erases the character to the left of the cursor.
Left Arrow or Ctrl+B	Moves the cursor one character to the left.
Right Arrow or Ctrl+F	Moves the cursor one character to the right.
Up Arrow or Ctrl+P	Recalls the commands in the history buffer, beginning with the most recent commands.

```
Router#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#
```

Basic Device Configuration

Basic Device Configuration

Device Names

- The first configuration command on any device should be to give it a unique hostname.
- By default, all devices are assigned a factory default name. For example, a Cisco IOS switch is "Switch."
- Guideline for naming devices:
 - Start with a letter
 - Contain no spaces
 - End with a letter or digit
 - Use only letters, digits, and dashes
 - Be less than 64 characters in length

```
Switch# configure terminal
Switch(config)# hostname Sw-Floor-1
Sw-Floor-1(config)#
```

Note: To return the switch to the default prompt, use the **no hostname** global config command.

Password Guidelines

- The use of weak or easily guessed passwords are a security concern.
- All networking devices should limit administrative access by securing privileged EXEC, user EXEC, and remote Telnet access with passwords. In addition, all passwords should be encrypted and legal notifications provided.

- Password Guidelines:

- Use passwords that are more than eight characters in length.
- Use a combination of upper and lowercase letters, numbers, special characters, and/or numeric sequences.
- Avoid using the same password for all devices.
- Do not use common words because they are easily guessed.



Note: Most of the labs in this course use simple passwords such as **cisco** or **class**. These passwords are considered weak and easily guessable and should be avoided in production environments.

Configure Passwords

Securing user EXEC mode access:

- First enter line console configuration mode using the **line console 0** command in global configuration mode.
- Next, specify the user EXEC mode password using the **password** *password* command.
- Finally, enable user EXEC access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line console 0
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Securing privileged EXEC mode access:

- First enter global configuration mode.
- Next, use the **enable** **secret** *password* command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# enable secret class
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

Configure Passwords (Cont.)

Securing VTY(Virtual Teletype) line access:

- First enter line VTY configuration mode using the **line vty 0 15** command in global configuration mode.
- Next, specify the VTY password using the **password password** command.
- Finally, enable VTY access using the **login** command.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# line vty 0 15
Sw-Floor-1(config-line)# password cisco
Sw-Floor-1(config-line)# login
Sw-Floor-1(config-line)# end
Sw-Floor-1#
```

Note: VTY lines enable remote access using Telnet or SSH to the device. Many Cisco switches support up to 16 VTY lines that are numbered 0 to 15.

Encrypt Passwords

- The startup-config and running-config files display most passwords in plaintext.
- To encrypt all plaintext passwords, use the **service password-encryption** global config command.
- Use the show running-config command to verify that the passwords on the device are now encrypted.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# service password-encryption
Sw-Floor-1(config)# exit
Sw-Floor-1#
```

```
Sw-Floor-1# show running-config
!
!
line con 0
password 7 094F471A1A0A
login
!
Line vty 0 4
Password 7 03095A0F034F38435B49150A1819
Login
!
!
end
```

Banner Messages

- A banner message is important to warn unauthorized personnel from attempting to access the device.
- To create a banner message of the day on a network device, use the **banner motd # the message of the day #** global config command.

Note: The “#” in the command syntax is called the delimiting character. It is entered before and after the message.

```
Sw-Floor-1# configure terminal
Sw-Floor-1(config)# banner motd #Authorized Access Only!#
```

The banner will be displayed on attempts to access the device.



```
Press RETURN to get started.
```

```
Authorized Access Only!
```

```
User Access Verification
```

```
Password:
```

Save Configurations

Save Configuration Files

- There are two system files that store the device configuration:
 - **startup-config** - This is the saved configuration file that is stored in NVRAM. It contains all the commands that will be used by the device upon startup or reboot. Flash does not lose its contents when the device is powered off.
 - **running-config** - This is stored in Random Access Memory (RAM). It reflects the current configuration. Modifying a running configuration affects the operation of a Cisco device immediately. RAM is volatile memory. It loses all of its content when the device is powered off or restarted.
 - To save changes made to the running configuration to the startup configuration file, use the **copy running-config startup-config** privileged EXEC mode command.

```
Router#show startup-config
Using 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```

```
Router#show running-config
Building configuration...

Current configuration : 624 bytes
!
version 15.4
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
```


Alter the Running Configurations

If changes made to the running config do not have the desired effect and the running-config has not yet been saved, you can restore the device to its previous configuration. To do this you can:

- Remove the changed commands individually.
- Reload the device using the **reload** command in privilege EXEC mode. *Note: This will cause the device to briefly go offline, leading to network downtime.*

```
Router# reload
Proceed with reload? [confirm]
Initializing Hardware ...
```

If the undesired changes were saved to the startup-config, it may be necessary to clear all the configurations using the **erase startup-config** command in privilege EXEC mode.

- After erasing the startup-config, reload the device to clear the running-config file from RAM.

```
Router# erase startup-config
Erasing the nvram filesystem will remove all configuration files! Continue? [confirm]
[OK]
Erase of nvram: complete
%SYS-7-NV_BLOCK_INIT: Initialized the geometry of nvram
Router#
```

LAN Security Concept

Endpoint Security

The news media commonly covers attacks on enterprise networks. Simply search the internet for “latest network attacks” to find up-to-date information on current attacks. Most likely, these attacks will involve one or more of the following:

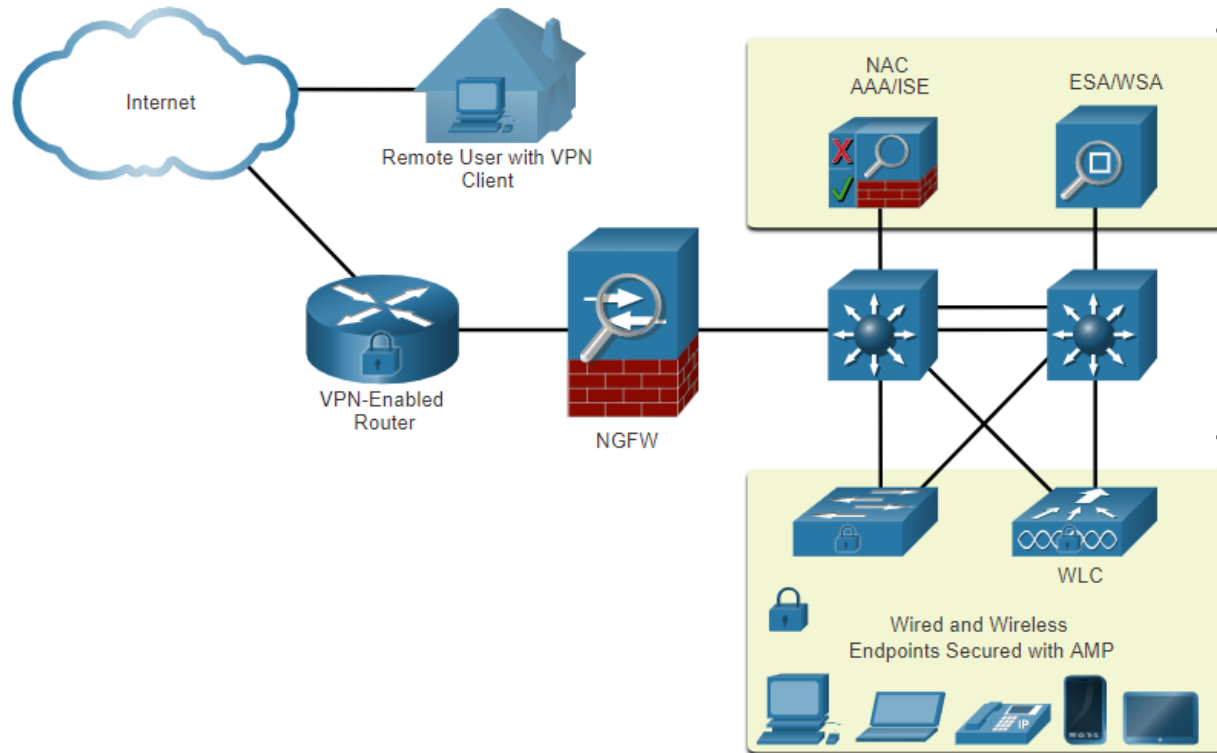
- **Distributed Denial of Service (DDoS)** – This is a coordinated attack from many devices, called zombies, with the intention of degrading or halting public access to an organization’s website and resources.
- **Data Breach** – This is an attack in which an organization’s data servers or hosts are compromised to steal confidential information.
- **Malware** – This is an attack in which an organization’s hosts are infected with malicious software that cause a variety of problems. For example, ransomware such as WannaCry encrypts the data on a host and locks access to it until a ransom is paid.

Network Security Devices

Various network security devices are required to protect the network perimeter from outside access. These devices could include the following:

- **Virtual Private Network (VPN)** enabled router - provides a secure connection to remote users across a public network and into the enterprise network. VPN services can be integrated into the firewall.
- **Next-Generation Firewall (NGFW)** - provides stateful packet inspection, application visibility and control, a next-generation intrusion prevention system (NGIPS), advanced malware protection (AMP), and URL filtering.
- **Network Access Control (NAC)** - includes authentication, authorization, and accounting (AAA) services. In larger enterprises, these services might be incorporated into an appliance that can manage access policies across a wide variety of users and device types. The Cisco Identity Services Engine (ISE) is an example of a NAC device.

Endpoint Protection



- Endpoints are hosts which commonly consist of laptops, desktops, servers, and IP phones, as well as employee-owned devices. Endpoints are particularly susceptible to malware-related attacks that originate through email or web browsing.
- Endpoints have typically used traditional host-based security features, such as antivirus/antimalware, host-based firewalls, and host-based intrusion prevention systems (HIPSs).
- Endpoints today are best protected by a combination of NAC, AMP software, an email security appliance (ESA), and a web security appliance (WSA).

Access Control using Authentication

- Many types of authentication can be performed on networking devices, and each method offers varying levels of security.
- The simplest method of remote access authentication is to configure a login and password combination on console, vty lines, and aux ports.
- SSH is a more secure form of remote access:
 - It requires a username and a password.
 - The username and password can be authenticated locally.
- The local database method has some limitations:
 - User accounts must be configured locally on each device which is not scalable.
 - The method provides no fallback authentication method.

```
R1(config)# line vty 0 4
R1(config-line)# password ci5c0
R1(config-line)# login
```

```
R1(config)# ip domain-name example.com
R1(config)# crypto key generate rsa general-keys modulus 2048
R1(config)# username Admin secret Str0ng3rPa55w0rd
R1(config)# ssh version 2
R1(config)# line vty 0 4
R1(config-line)# transport input ssh
R1(config-line)# login local
```

AAA Components

- AAA stands for Authentication, Authorization, and Accounting, and provides the primary framework to set up access control on a network device.
- AAA is a way to control who is permitted to access a network (authenticate), what they can do while they are there (authorize), and to audit what actions they performed while accessing the network (accounting).
 - ❑ Local and server-based are two common methods of implementing AAA authentication.
 - Local AAA Authentication
 - Server-Based AAA Authentication
 - ❑ AAA authorization is automatic and does not require users to perform additional steps after authentication. Authorization governs what users can and cannot do on the network after they are authenticated.
 - ❑ AAA accounting collects and reports usage data. This data can be used for such purposes as auditing or billing. The collected data might include the start and stop connection times, executed commands, number of packets, and number of bytes.

Switch Port Security

- Physically unsecured switched can have unauthorised devices attached.
- Potential hacker can pull out ethernet cable and plug in another device.
- Solution:
 - Administratively shut down unused switch ports
 - Filter MAC addresses at switch port level (configure the switch ports to allow frames from authorised devices only using MAC address)

Switch Port Security: Secure Unused Ports

Layer 2 attacks are some of the easiest for hackers to deploy but these threats can also be mitigated with some common Layer 2 solutions.

- All switch ports (interfaces) should be secured before the switch is deployed for production use. How a port is secured depends on its function.
- A simple method that many administrators use to help secure the network from unauthorized access is to disable all unused ports on a switch. Navigate to each unused port and issue the Cisco IOS **shutdown** command. If a port must be reactivated at a later time, it can be enabled with the **no shutdown** command.
- To configure a range of ports, use the **interface range** command.

```
Switch(config)# interface range type module/first-number - last-number
```

Switch Port Security

- Shut down switch port

```
home-switch(config)#interface fa2/1
home-switch(config-if)#shutdown
```

```
%LINK-5-CHANGED: Interface FastEthernet4/1, changed state to administratively down
home-switch(config-if)#exit
home-switch#
```

```
S1# show port-security interface f0/1
Port Security           : Enabled
Port Status              : Secure-shutdown
Violation Mode           : Shutdown
Aging Time               : 0 mins
Aging Type               : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses    : 1
Total MAC Addresses      : 0
Configured MAC Addresses : 0
Sticky MAC Addresses     : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
S1#
```

- Restrict Access to Port

```
home-switch(config)#interface fa0/1
home-switch(config-if)#switchport mode access
home-switch(config-if)#switchport port-security
home-switch(config-if)#switchport port-security mac-address sticky
home-switch(config-if)#switchport port-security maximum 1
home-switch(config-if)#switchport port-security violation shutdown
```

```
S1(config)# interface f0/1
S1(config-if)# switchport port-security maximum ?
<1-8192> Maximum addresses
S1(config-if)# switchport port-security maximum
```

Switch Port Security

- After port security is enabled, other port security specifics can be configured, as shown in the example.

```
S1(config-if)# switchport port-security ?  
  aging      Port-security aging commands  
  mac-address Secure mac address  
  maximum    Max secure addresses  
  violation   Security violation mode  
  <cr>  
S1(config-if)# switchport port-security
```

Limit and Learn MAC Addresses

The switch can be configured to learn about MAC addresses on a secure port in one of three ways:

- **Manually Configured:** The administrator manually configures a static MAC address(es) by using the following command for each secure MAC address on the port:

```
Switch(config-if)# switchport port-security mac-address mac-address
```

- **Dynamically Learned:** When the switchport port-security command is entered, the current source MAC for the device connected to the port is automatically secured but is not added to the running configuration. If the switch is rebooted, the port will have to re-learn the device's MAC address.
- **Dynamically Learned – Sticky:** The administrator can enable the switch to dynamically learn the MAC address and “stick” them to the running configuration by using the following command:

```
Switch(config-if)# switchport port-security mac-address sticky
```

Saving the running configuration will commit the dynamically learned MAC address to NVRAM.

Limit and Learn MAC Addresses

The example demonstrates a complete port security configuration for FastEthernet 0/1.

- The administrator specifies a maximum of 4 MAC addresses, manually configures one secure MAC address, and then configures the port to dynamically learn additional secure MAC addresses up to the 4 secure MAC address maximum.
- Use the show port-security interface and the show port-security address command to verify the configuration.

```
S1(config)# interface fa0/1
S1(config-if)# switchport mode access
S1(config-if)# switchport port-security
S1(config-if)# switchport port-security maximum 4
S1(config-if)# switchport port-security mac-address aaaa.bbbb.1234
S1(config-if)# switchport port-security mac-address sticky
S1(config-if)# end
S1# show port-security interface fa0/1
Port Security          : Enabled
Port Status            : Secure-up
Violation Mode         : Shutdown
Aging Time             : 0 mins
Aging Type             : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses  : 4
Total MAC Addresses    : 1
Configured MAC Addresses : 1
Sticky MAC Addresses   : 0
Last Source Address:Vlan : 0000.0000.0000:0
Security Violation Count : 0
S1# show port-security address
                        Secure Mac Address Table
-----
Vlan    Mac Address      Type                Ports    Remaining Age
      (mins)
-----
      1    aaaa.bbbb.1234  SecureConfigured    Fa0/1    -
-----
Total Addresses in System (excluding one mac per port)  : 0
Max Addresses limit in System (excluding one mac per port) : 8192

S1#
```